

Lecture 2: Asymptotic Theory in Depth

MLE Consistency, Asymptotic Normality, Delta Method,
and Where Asymptotics Break Down

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1 Recap and Roadmap

In Lecture 1 we saw that the CLT provides approximate normality of $\bar{X}_n$, but the approximation error for finite n is uncontrolled. Today we go deeper into the asymptotic paradigm, studying its core concept — the *Maximum Likelihood Estimator* (MLE) — and then examining, with equal care, the conditions under which the theory fails.

Lecture 2 roadmap:

1. **MLE consistency:** conditions and proof via WLLN.

2. **Asymptotic normality and Fisher information:** the $\sqrt{n}$ -rate and the Delta method.
3. **Where asymptotics break down:** misspecification, heavy tails, high dimensions — and what to do.

Connecting Thread

Every positive result in this lecture rests on a list of assumptions. By the end of the lecture, we will have identified exactly which assumption can be relaxed and what the cost is — motivating the non-asymptotic approach of Lectures 3 and 4.

2 Setup: The Statistical Model and the MLE

Definition 2.1 — Parametric Model

A **parametric statistical model** is a family of distributions $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ indexed by a parameter θ in an open subset $\Theta \subseteq \mathbb{R}^d$. We assume each P_θ has a density $p(\cdot; \theta)$: a probability density function if the data are continuous, or a probability mass function if the data are discrete.

Definition 2.2 — Log-likelihood and MLE

Given observations $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p(\cdot; \theta_0)$, the **log-likelihood** is

$$\ell_n(\theta) = \sum_{i=1}^n \log p(X_i; \theta).$$

The **Maximum Likelihood Estimator** is

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} \ell_n(\theta),$$

assuming the maximum exists and is unique.

Example 2.3 — Gaussian MLE

Let $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$. The log-likelihood is

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Setting partial derivatives to zero gives

$$\hat{\mu}_n = \bar{X}_n, \quad \hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Note: $\hat{\sigma}_n^2$ is biased (divides by n , not $n - 1$), but consistent.

Example 2.4 — Bernoulli MLE

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $\theta = p \in (0, 1)$; here the density is taken with respect to counting measure, $p(x; p) = p^x(1 - p)^{1-x}$ for $x \in \{0, 1\}$. Writing $S_n := \sum_{i=1}^n X_i$, the log-likelihood is

$$\ell_n(p) = \sum_{i=1}^n [X_i \log p + (1 - X_i) \log(1 - p)] = S_n \log p + (n - S_n) \log(1 - p).$$

Setting $\ell'_n(p) = \frac{S_n}{p} - \frac{n - S_n}{1 - p} = 0$ gives

$$\hat{p}_n = \frac{S_n}{n} = \bar{X}_n,$$

and $\ell''_n(p) < 0$ confirms it is the maximizer. The MLE is unbiased ($\mathbb{E}[\hat{p}_n] = p$) and consistent ($\hat{p}_n \xrightarrow{p} p$ by the WLLN). Since $S_n \sim \text{Binomial}(n, p)$, this is equally the MLE from a single $\text{Binomial}(n, p)$ observation.

3 MLE Consistency

3.1 The key idea: M-estimation

The MLE belongs to the general class of **M-estimators**: estimators defined as the maximizer (or minimizer) of an objective function of the form $\frac{1}{n} \sum_{i=1}^n m(X_i, \theta)$. For the MLE, $m(x, \theta) = \log p(x; \theta)$.

The core idea is: by the WLLN,

$$\frac{1}{n} \ell_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log p(X_i; \theta) \xrightarrow{p} \mathbb{E}_{\theta_0}[\log p(X_1; \theta)] \quad \text{for each fixed } \theta.$$

The limit $M(\theta) := \mathbb{E}_{\theta_0}[\log p(X_1; \theta)]$ is maximized at $\theta = \theta_0$ by the information inequality.

Lemma 3.1 (Information Inequality / Gibbs's Inequality). *For any $\theta \in \Theta$,*

$$\mathbb{E}_{\theta_0}[\log p(X_1; \theta)] \leq \mathbb{E}_{\theta_0}[\log p(X_1; \theta_0)],$$

with equality if and only if $p(\cdot; \theta) = p(\cdot; \theta_0)$ a.e.

Proof. By Jensen's inequality applied to the concave function $\log$:

$$\mathbb{E}_{\theta_0} \left[\log \frac{p(X_1; \theta)}{p(X_1; \theta_0)} \right] \leq \log \mathbb{E}_{\theta_0} \left[\frac{p(X_1; \theta)}{p(X_1; \theta_0)} \right] = \log \int p(x; \theta) dx = \log 1 = 0.$$

The quantity on the left is $M(\theta) - M(\theta_0)$, so $M(\theta) \leq M(\theta_0)$. $\square$

Remark — KL Divergence

The inequality in the proof came from bounding $\mathbb{E}_{\theta_0} \left[\log \frac{p(X_1; \theta)}{p(X_1; \theta_0)} \right]$. Reading the two sides directly gives

$$M(\theta_0) - M(\theta) = \mathbb{E}_{\theta_0} \left[\log \frac{p(X_1; \theta_0)}{p(X_1; \theta)} \right] =: \text{KL}(P_{\theta_0} \| P_{\theta}) \geq 0,$$

the Kullback–Leibler divergence, which is ≥ 0 by Gibbs’s inequality (Lemma 3.1) with equality iff $P_{\theta} = P_{\theta_0}$ a.e. Rearranging,

$$M(\theta) = \underbrace{M(\theta_0)}_{\text{const in } \theta} - \text{KL}(P_{\theta_0} \| P_{\theta}), \quad M(\theta_0) = \mathbb{E}_{\theta_0} [\log p(X_1; \theta_0)] = -H(P_{\theta_0}),$$

where the constant $M(\theta_0)$ is the *negative (differential) entropy* of P_{θ_0} : it depends on the data-generating θ_0 but *not* on the argument θ , so it plays no role in locating the maximizer.

Consistency of the MLE is thus fundamentally a statement that the normalized log-likelihood $\frac{1}{n} \ell_n(\theta) \xrightarrow{P} M(\theta) = \text{const} - \text{KL}(P_{\theta_0} \| P_{\theta})$. Since $\text{KL} \geq 0$ vanishes only when $P_{\theta} = P_{\theta_0}$, and *identifiability* ($P_{\theta} = P_{\theta_0} \Rightarrow \theta = \theta_0$) promotes this to a statement about parameters, the limit is *uniquely* maximized at $\theta = \theta_0$. Maximizing the likelihood is, in the limit, the same as minimizing KL divergence to the truth.

3.2 Consistency theorem

Theorem 3.1 — Consistency of the MLE

Suppose:

- (A1) **Identifiability:** $\theta \neq \theta' \Rightarrow P_{\theta} \neq P_{\theta'}$.
- (A2) **Compact parameter space:** Θ is compact.
- (A3) **Uniform convergence:** $\sup_{\theta \in \Theta} \left| \frac{1}{n} \ell_n(\theta) - M(\theta) \right| \xrightarrow{P} 0$.
- (A4) **Continuity:** $M(\theta)$ is continuous on Θ .

Then $\hat{\theta}_n \xrightarrow{P} \theta_0$.

Proof idea. The MLE maximizes $\frac{1}{n} \ell_n$, which by (A3) is uniformly close to M ; so its maximizer $\hat{\theta}_n$ must be close to the maximizer θ_0 of M . We skip the technical details, which require a compactness argument and the continuity of M . See van der Vaart, *Asymptotic Statistics* (Theorem 5.7) for details. $\square$

4 Asymptotic Normality and Fisher Information

4.1 Fisher information

Definition 4.1 — Score Function and Fisher Information

The **score function** is

$$s(x; \theta) = \nabla_{\theta} \log p(x; \theta).$$

The **Fisher information matrix** at θ is

$$\mathcal{I}(\theta) = \mathbb{E}_{\theta} [s(X; \theta) s(X; \theta)^{\top}] \in \mathbb{R}^{d \times d}.$$

Under regularity (interchange of differentiation and integration), $\mathbb{E}_{\theta}[s(X; \theta)] = 0$ and equivalently

$$\mathcal{I}(\theta) = -\mathbb{E}_{\theta} [\nabla_{\theta}^2 \log p(X; \theta)].$$

Example 4.2 — Fisher Information for $\mathcal{N}(\mu, \sigma^2)$

With $\theta = \mu$ (known σ^2): $\log p(x; \mu) = -\frac{(x-\mu)^2}{2\sigma^2} + \text{const}$, so $s(x; \mu) = (x - \mu)/\sigma^2$ and

$$\mathcal{I}(\mu) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right] = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

More information (smaller σ^2) means easier estimation — as expected.

4.2 Asymptotic normality of the MLE

Theorem 4.3 — Asymptotic Normality of the MLE

Suppose (A1)–(A4) hold and additionally:

(A5) **Interior:** $\theta_0 \in \text{int}(\Theta)$.

(A6) **Twice differentiability:** $\log p(x; \theta)$ is twice continuously differentiable in θ for a.e. x .

(A7) **Dominated second derivative:** $\mathbb{E}_{\theta_0} [\sup_{\theta} \|\nabla_{\theta}^2 \log p(X_1; \theta)\|] < \infty$.

(A8) **Non-singular Fisher information:** $\mathcal{I}(\theta_0)$ is positive definite.

Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}).$$

Proof. Since $\hat{\theta}_n$ maximizes ℓ_n and $\theta_0 \in \text{int}(\Theta)$, the first-order condition gives $\nabla_{\theta} \ell_n(\hat{\theta}_n) = 0$. Taylor-expand around θ_0 :

$$0 = \nabla \ell_n(\hat{\theta}_n) = \nabla \ell_n(\theta_0) + \nabla^2 \ell_n(\tilde{\theta}_n) (\hat{\theta}_n - \theta_0),$$

where $\tilde{\theta}_n$ lies between $\hat{\theta}_n$ and θ_0 . Rearranging:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = - \left[\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) \right]^{-1} \frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0).$$

Numerator. The score vector satisfies $\mathbb{E}_{\theta_0}[s(X_i; \theta_0)] = 0$ and $\text{Var}(s(X_i; \theta_0)) = \mathcal{I}(\theta_0)$. By the CLT:

$$\frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n s(X_i; \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)).$$

Denominator. By the WLLN and (A7),

$$\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) \xrightarrow{p} \mathbb{E}_{\theta_0}[\nabla_{\theta}^2 \log p(X_1; \theta_0)] = -\mathcal{I}(\theta_0).$$

Slutsky's theorem (if $A_n \xrightarrow{p} A$ and $B_n \xrightarrow{d} B$, then $A_n B_n \xrightarrow{d} AB$) then gives

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{I}(\theta_0)^{-1} \cdot \mathcal{N}(0, \mathcal{I}(\theta_0)) = \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}). \quad \square$$

Remark — The Cramér–Rao Lower Bound

The Cramér–Rao theorem states that for any unbiased estimator T_n of θ_0 ,

$$\text{Var}(T_n) \geq \frac{1}{n} \mathcal{I}(\theta_0)^{-1}.$$

Theorem 4.3 shows the MLE *achieves* this bound asymptotically: the MLE is **asymptotically efficient**. No regular estimator can do better at first order.

4.3 The Delta method

A common task is estimating $g(\theta_0)$ for a smooth function $g : \mathbb{R}^d \rightarrow \mathbb{R}^k$. The Delta method provides the asymptotic distribution.

Theorem 4.4 — Delta Method

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is continuously differentiable at θ with Jacobian $\nabla g(\theta)$. Then

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, \nabla g(\theta)^\top \Sigma \nabla g(\theta)).$$

Proof. By first-order Taylor expansion at θ : $g(T_n) - g(\theta) = \nabla g(\theta)^\top (T_n - \theta) + o(\|T_n - \theta\|)$. Since $T_n \xrightarrow{p} \theta$ (by $\sqrt{n}$ -consistency), the remainder is $o_p(1/\sqrt{n})$, so

$$\sqrt{n}(g(T_n) - g(\theta)) = \nabla g(\theta)^\top \cdot \sqrt{n}(T_n - \theta) + o_p(1) \xrightarrow{d} \nabla g(\theta)^\top \mathcal{N}(0, \Sigma) = \mathcal{N}(0, \nabla g(\theta)^\top \Sigma \nabla g(\theta)).$$

□

Example 4.5 — CI for the Odds Ratio

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$. We want a CI for the odds $g(p) = p/(1-p)$. We have $\hat{p}_n = \bar{X}_n$ with $\sqrt{n}(\hat{p}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$. The derivative is $g'(p) = 1/(1-p)^2$, so by the Delta method:

$$\sqrt{n} \left(\frac{\hat{p}_n}{1 - \hat{p}_n} - \frac{p}{1 - p} \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{p(1-p)}{(1-p)^4} \right) = \mathcal{N} \left(0, \frac{p}{(1-p)^3} \right).$$

An approximate 95% CI for the odds is $\hat{p}_n/(1 - \hat{p}_n) \pm 1.96\sqrt{\hat{p}_n/(n(1 - \hat{p}_n)^3)}$.

5 Where Asymptotics Break Down

The results of Sections 3–4 are powerful, but each rests on assumptions that can fail in practice. We examine three failure modes and use them to motivate the non-asymptotic perspective.

5.1 Failure Mode 1: Model misspecification

Theorem 4.3 assumes the data truly come from $p(\cdot; \theta_0)$. What if the true distribution Q is not in $\{P_\theta\}$?

Theorem 5.1 — MLE under Misspecification (White, 1982)

Suppose the data are i.i.d. from $Q \notin \{P_\theta\}$. Under regularity conditions, the MLE converges to the **pseudo-true parameter**:

$$\hat{\theta}_n \xrightarrow{p} \theta^* := \arg \min_{\theta \in \Theta} \text{KL}(Q \| P_\theta).$$

Moreover,

$$\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, A(\theta^*)^{-1} B(\theta^*) A(\theta^*)^{-1}),$$

where $A(\theta^*) = -\mathbb{E}_Q[\nabla^2 \log p(X; \theta^*)]$ and $B(\theta^*) = \text{Var}_Q(\nabla \log p(X; \theta^*))$.

Remark — The Sandwich Variance

The asymptotic variance $A^{-1}BA^{-1}$ is called the **sandwich estimator**. Under correct specification, $A = B = \mathcal{I}(\theta_0)$, so $A^{-1}BA^{-1} = \mathcal{I}(\theta_0)^{-1}$ — recovering Theorem 4.3. Under misspecification, $A \neq B$, and using $\mathcal{I}(\hat{\theta}_n)^{-1}$ as the variance underestimates the true uncertainty. **Standard errors based on the Fisher information are incorrect under misspecification.**

5.2 Failure Mode 2: Heavy tails**Remark — CLT Fails for Heavy-Tailed Distributions**

If $\text{Var}(X_1) = \infty$ (e.g., $X_1 \sim \text{Pareto}(\alpha)$ with $\alpha \leq 2$), the classical CLT does not apply. The normalized sum $n^{-1/\alpha} \sum_{i=1}^n X_i$ converges to a **stable distribution** (not Gaussian) with α -stable index.

Example 5.3 — Sample Mean of Pareto Data

Let $X_i \stackrel{\text{iid}}{\sim} \text{Pareto}(\alpha = 1.5)$ with scale $x_m = 1$, so $\mathbb{E}[X_i] = \frac{\alpha x_m}{\alpha - 1} = 3$ but $\text{Var}(X_i) = \infty$ (the second moment diverges since $\alpha \leq 2$). The CLT does not apply: the correctly normalized mean $n^{1-1/\alpha}(\bar{X}_n - \mu)$ converges to an α -stable law, not a Gaussian, so the usual $\sqrt{n}$ scaling and the Gaussian CI are both wrong.

Figure 1 shows a simulation with $R = 3 \times 10^5$ replications. Panel (a): the histogram of $\sqrt{n}(\bar{X}_n - \mu)$ at $n = 1000$ is strongly right-skewed — its bulk sits *below* 0 (the many small draws pull $\bar{X}_n$ under μ) while a heavy right tail comes from rare enormous draws (about 1.4% of samples land beyond the plotted range). Panel (b): the nominal 95% CI $\bar{X}_n \pm 1.96 \hat{\sigma}/\sqrt{n}$ never reaches its target — actual coverage is only $\approx 70\%$ at $n = 100$ and creeps to just $\approx 79\%$ at $n = 10^4$. The misses are almost entirely one-sided (the interval falls short of μ from below), because $\hat{\sigma}$ is itself unstable when the variance is infinite.

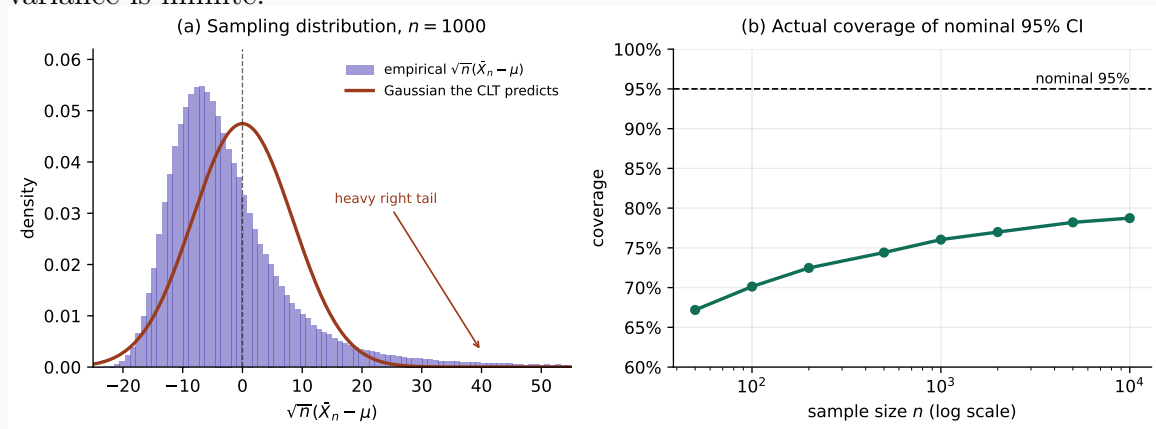


Figure 1: Failure of the CLT for Pareto(1.5) data. (a) Sampling distribution of $\sqrt{n}(\bar{X}_n - \mu)$ vs. the Gaussian the CLT would predict; (b) coverage of the nominal 95% CI, which plateaus well below 95% as n grows.

In contrast, non-asymptotic bounds using only $\mathbb{E}[|X|^q] < \infty$ for some $q \in (1, 2]$ (e.g. median-of-means or truncation-based estimators) provide valid, if wider, intervals with guaranteed coverage at every n — a preview of Lectures 3–4.

5.3 Failure Mode 3: High dimensions ($p \approx n$ or $p \gg n$)

Remark — High-Dimensional Breakdown

Classical asymptotic theory assumes d (dimension of θ) is fixed as $n \rightarrow \infty$. When $d = d_n \rightarrow \infty$ with n , several pathologies arise:

- **MLE can be inconsistent.** In logistic regression with $p/n \rightarrow \kappa \in (0, 1)$, the MLE is inconsistent: $\hat{\theta}_n \xrightarrow{p} c(\kappa) \theta_0$ for a bias factor $c(\kappa) > 1$ (Sur & Candès, 2019).
- **Covariance estimation breaks down.** The sample covariance $\hat{\Sigma}_n$ is not a consistent estimator of Σ in operator norm when $p/n \not\rightarrow 0$. The Marchenko–Pastur law describes the bulk eigenvalue distribution of $\hat{\Sigma}_n$ when $p/n \rightarrow \gamma \in (0, \infty)$.
- **Confidence intervals have wrong coverage.** Even if point estimates are good, the χ^2 approximation to the likelihood ratio statistic breaks down.

Example 5.4 — Sample Covariance in High Dimensions

Draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_p)$, so the true covariance is $\Sigma = I_p$ and *every* population eigenvalue equals 1. Form the sample covariance $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top$. Even a perfect estimator should return eigenvalues near 1; Figure 2 shows what actually happens.

Eigenvalues spread out (panel a). At $p/n = 0.5$ ($p = 250, n = 500$) the empirical eigenvalues of $\hat{\Sigma}_n$ fill the whole interval $[(1 - \sqrt{\gamma})^2, (1 + \sqrt{\gamma})^2] \approx [0.09, 2.91]$, tracing the **Marchenko–Pastur** density with $\gamma = 0.5$ — not a spike at 1. The largest sample eigenvalue is nearly $3\times$ the smallest, purely an artifact of dimension.

No consistency in operator norm (panel b). Consequently $\|\hat{\Sigma}_n - \Sigma\|_{\text{op}} \rightarrow (1 + \sqrt{\gamma})^2 - 1 \approx 1.91$ and *does not vanish* as $n \rightarrow \infty$ when p/n is held at 0.5. Only when the dimension is fixed ($p = 10$) does the classical $O(\sqrt{p/n})$ decay kick in.

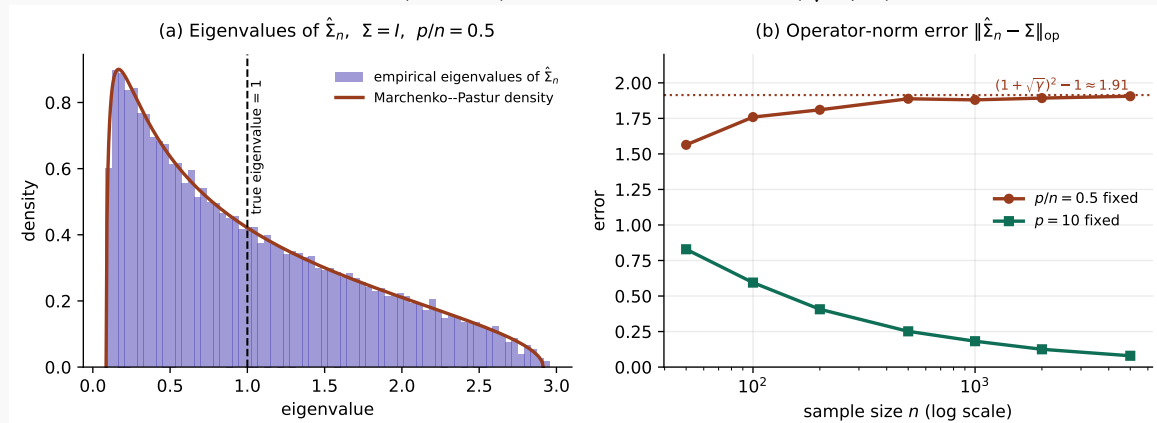


Figure 2: Breakdown of sample-covariance estimation with $\Sigma = I$. (a) Eigenvalues of $\hat{\Sigma}_n$ at $p/n = 0.5$ follow the Marchenko–Pastur law, spread far from the true value 1; (b) operator-norm error stays ≈ 1.91 under proportional scaling ($p/n = 0.5$) but decays to 0 when p is fixed.

The lesson: entrywise, $\hat{\Sigma}_n$ is still consistent (each entry is a sample mean), but as a *matrix* it is not — spectral functionals, PCA, and Wald-type tests built on $\hat{\Sigma}_n^{-1}$ all inherit this error. This is a regime where non-asymptotic, dimension-explicit theory (Lectures 3–4) is the primary tool.

5.4 Summary of failure modes

Failure mode	What breaks	Remedy
Misspecification	CI coverage, efficiency	Sandwich SE; robust estimators
Heavy tails ($\text{Var} = \infty$)	CLT, rate $\sqrt{n}$	Stable-law CLT; non-asymptotic bounds with truncation
High dimensions ($p \asymp n$)	Consistency, χ^2 approx.	Regularization; RMT corrections; non-asymptotic guarantees

The Key Takeaway for This Course

The asymptotic paradigm gives sharp, interpretable results *when its assumptions hold*. The three failure modes above are not edge cases — they arise routinely in modern statistics and machine learning. **Non-asymptotic theory** does not require $n \rightarrow \infty$, does not need the model to be correctly specified (distribution-free bounds), and scales naturally to high dimensions. This is why Lectures 3 and 4 develop it in depth.

6 Summary

1. The **MLE is consistent** under identifiability, compactness, and uniform convergence of the log-likelihood. The proof uses the WLLN and the fact that θ_0 uniquely maximizes $M(\theta)$ via the information inequality ($\text{KL} \geq 0$).
2. The **MLE is asymptotically normal** with variance $\mathcal{I}(\theta_0)^{-1}$, achieving the Cramér–Rao lower bound. The proof combines a Taylor expansion of the score with the CLT and Slutsky’s theorem.

3. The **Delta method** extends asymptotic normality to smooth functionals $g(\theta_0)$ via a first-order Taylor expansion.
4. **Three failure modes** break the asymptotic guarantees: model misspecification (sandwich variance needed), heavy tails (CLT breaks), and high dimensions (MLE inconsistency, χ^2 approximation fails).
5. These failure modes motivate non-asymptotic theory, which provides guarantees that are *distribution-free*, *valid for all n* , and *dimension-aware* — at the cost of potentially looser bounds.

Looking ahead. Lecture 3 develops the machinery of concentration inequalities — Hoeffding, Bernstein, and the sub-Gaussian family — and directly compares their guarantees with the asymptotic results derived today.

7 Exercises

Exercise 7.1 (MLE for Poisson). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$.

- (a) Compute $\hat{\lambda}_n$ and verify consistency directly.
- (b) Compute $\mathcal{I}(\lambda)$ and state the asymptotic distribution of $\hat{\lambda}_n$.
- (c) Use the Delta method to find the asymptotic distribution of $\log \hat{\lambda}_n$.

Exercise 7.2 (Score and Fisher information). For a model $p(x; \theta) = \theta e^{-\theta x}$ ($x > 0, \theta > 0$):

- (a) Show $\mathbb{E}_\theta[s(X; \theta)] = 0$ by direct computation.
- (b) Compute $\mathcal{I}(\theta)$ using both definitions in Definition 4.1 and verify they agree.
- (c) What is the asymptotic variance of $\hat{\theta}_n$?

Exercise 7.3 (Sandwich variance). Suppose $X_i \stackrel{\text{iid}}{\sim} Q$ where Q has mean μ and variance σ_Q^2 , but we fit the model $p(x; \mu) = \mathcal{N}(x; \mu, 1)$ (known unit variance).

- (a) Find the pseudo-true parameter θ^* .
- (b) Compute the matrices $A(\theta^*)$ and $B(\theta^*)$ from Theorem 5.1.
- (c) Show that the sandwich variance reduces to σ_Q^2/n , while the naive Fisher-based variance is $1/n$. When are these equal?

Exercise 7.4 (Failure in high dimensions). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_p)$ and $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top$ the sample covariance.

- (a) Show $\mathbb{E}[\hat{\Sigma}_n] = I_p$ (unbiased).
- (b) Show $\|\hat{\Sigma}_n - I_p\|_F^2 = O(p^2/n)$ in expectation, so $\hat{\Sigma}_n$ is consistent in Frobenius norm only if $p^2/n \rightarrow 0$.

(c) What does this imply for the regime $p \asymp n$?

Exercise 7.5 (Cramér–Rao via Fisher information). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known.

(a) Compute $\mathcal{I}(\mu)$ and the Cramér–Rao lower bound for estimating μ .

(b) Verify that $\bar{X}_n$ achieves the bound (exactly, not just asymptotically).

(c) Now suppose σ^2 is unknown. Is $\bar{X}_n$ still efficient?

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